Progressive exercises

E1 · Practice. Let X be a random variable such that $E(X) = 3$ and $V(X) = 2$. Let $Y = 4X - 1$. Calculate the expected value, the variance and the standard deviation of Y.

E2 · Practice. The random variables X and Y are independent and satisfy $E(X) = 2$, $E(Y) = 5$, $V(X) = 1$ and $V(Y) = 4$. Calculate $E(2X - Y)$ and $V(2X - Y)$.

E3 · Practice. Consider 64 independent random variables with the same distribution, with expected value 10 and standard deviation 4. Calculate the expected value and the standard deviation of their mean.

E4 · Practice. Let X be a random variable such that $E(X) = 100$ and $V(X) = 9$. Using the Bienaymé–Chebyshev inequality, give an upper bound for $P(|X - 100| \geq 6)$.

E5 · Practice. Let M_n be the mean of n independent random variables with the same distribution, with expected value μ and variance 4. Determine a size n sufficient for $P(|M_n - \mu| < 0.2) \geq 0.95$.

Textbook or past-paper work

Baccalauréat, Metropolitan France, 17 June 2025: exercise 1, question 6 (p. 3). At home after S77. Read the context of question 5 and use the expected value and variance values accepted in question 6. The paper is available here: official past paper. Use the data and instructions of this past paper.